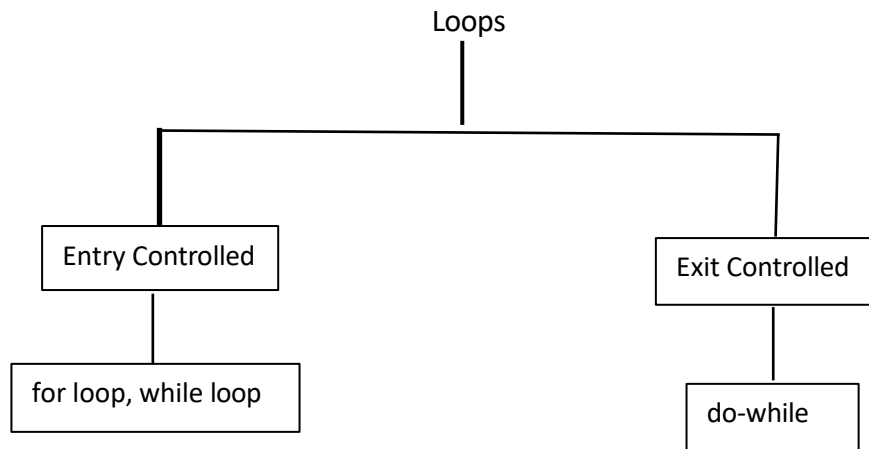


Iterations (loops)

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A loop in Java is a control flow statement that allows code to be executed repeatedly based on a condition. Each type of loop executes a block of code multiple times until a specified condition is met. This allows for repetitive tasks to be automated efficiently.



Entry Controlled Loops: Loops in which condition is checked first and code is executed only if condition is true are called Entry Controlled Loops. Example: for loop and while loop

Exit Controlled Loops: In Exist controlled loops condition is checked after executing the code. here the loops runs at least once even if the condition is false. Example: do while loop

** In this document we will discuss about for loop.

for loop Syntax:

```
for(initialization; condition; update statement){  
    //code  
}
```

① → ② → ③ → ④ → ②

Example 1:

```
for(int i=1; i<=5; i++){  
    System.out.print(i);  
}
```

Output: 1 2 3 4 5

Example 2:

```
for(int i=1; i<=10; i+=2){  
    System.out.print(i);  
}
```

Output: 1 3 5 7 9

Example 3:

```
for(int i=10; i>=1; i--){  
    System.out.println(i);  
}
```

Output: 10 9 8 7 6 5 4 3 2 1

Example 3:

```
for(double i=0.0; i<=2.0; i+=  
    0.5){  
    System.out.print(i);  
}
```

Output: 0.0 0.5 1.0 1.5 2.0

To find how many times the loop runs: (NOT WHEN JUMP STATEMENTS ARE USED)

```
for (int i=1; i<=10; i++){  
    System.out.println(i);  
}
```

We can clearly see that the loop runs for 10 times

We can also find this using a formula:

```
for(int i=a; i<=b; i+=k){  
}
```

Number of iterations = $\left\lfloor \frac{b-a}{k} \right\rfloor + 1$

Applying this formula to the loop above :

Number of iterations = $\left\lfloor \frac{10-1}{1} \right\rfloor + 1 = 10$

****Important**

```
int a=5, b=5;  
for(int i=1; i<= 10; i++){  
    a++;  
    b++;  
}
```

After this loop: a and b both are 10

```
int a=5, b=5;  
for(int i=1; i<= 10; i++){  
    a++;  
    b++;  
}
```

Here, after the loop, a is 10 but b is 6

As no compound statement { } is written- only
a++ is inside loop's body and b++ is outside loop

```
int i, a=5;  
for(i=1; i<=5; i++);  
a++;
```

as the loop here is terminated using ; it has no body, this means a++ is outside the loop and runs only once like any normal line of code. It is also known as NULL Loop

after this loop: a is 6 and i is also 6 (loop runs only 5 times, but the loop condition fails when i=6)

Sequence

A sequence is an ordered list of that follow a specific rule or pattern.

Example 1: 1 2 3 4 5

Example 2: 1 3 5 7 9

We can print a sequence with for loop.

Example 1:

1, 2, 4, 8, 16, 32, 64

2^0 2^1 2^2 2^3 2^4 2^5 2^6

The term looks like: 2^i [0,6]

```
for( int i= 0; i<=6; i++){
```

```
double k= Math.pow(2,i);
```

```
System.out.println(k+ ",");
```

```
}
```

**** For sequence, We must write the printing code inside the loop block.**

Example 2:

2 9 28 65 126

1^3+1 2^3+1 3^3+1 4^3+1 5^3+1

the term looks like: i^3+1 [1,5]

```
for(int i=1; i<=5; i++){
```

```
double k= Math.pow(i,3)+1;
```

```
System.out.println(k);
```

Series

****We need to understand two symbols in order to gain better understanding of sum of series.**

Capital Sigma (Σ) and Capital PI (Π)

Sigma (Summation)

Σ This symbol is used to represent summation. There are two types of it – Limited and Limitless. We are interested in limited Sigma for our purpose.

The limited Sigma has two limits, upper limit and lower limit and a function to find sum of.

represented as $\sum_{\text{lower limit}}^{\text{Upper limit}} f$, It finds sum of all f from lower limit to upper limit with an update of +1

Example 1:

$$1 + 2 + 3 + 4 + \dots + n = \sum_{i=1}^n i$$

With Loop:

```
int s=0;
for(int i=1; i<=n; i++){
    s+=i;
}
```

A series is the sum of the terms of a sequence.

Example 1: $S = 1 + 2 + 3 + 4 + 5$

Example 2: $S = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}$

We can find S (sum of sequence or series) using loop. Here's how:

Example 1:

$$S = 1 + 2 + 3 + 4 + 5$$

$$+ + \sum_{i=1}^5 i$$

we need to create a variable to store the sum (say s)

```
int s=0;
```

```
for(int i=1; i<=5; i++){
```

```
    s+= i;
```

```
}
```

```
System.out.println(s);
```

Example 2:

$$S = \frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{9} + \frac{1}{11} + \frac{1}{13}$$
$$+= 2 \sum_{i=1}^{13} 1/i$$

```
int s=0;
```

```
for(int i=1; i<=13; i+=2){
```

```
    s+= 1/i;
```

```
}
```

```
System.out.println(s);
```

***** In Series, always print S outside loop block***

PI (Pruduct)

**** Same goes for PI, but it gives us the product.**

$$+ + \prod_{i=1}^n i = 1 * 2 * 3 * 4 * 5 * * n$$

**** this is same as n!**

With Loop:

```
int p=1;
for(int i=1; i<=n; i++){
    p*=i;
}
```